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# Data Analytics

## 4-Month Training Syllabus



### **DURATION**

4 Months (16 Weeks)



### **CLASS TIMING**

2 Hours per Day



### **TOOLS**

Excel, SQL, Tableau, Power BI

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# Program Overview



## DURATION

**4 Months** (16 Weeks)



## CLASS TIMING

**2 Hours per Day**



## TOOLS COVERED

Excel

SQL

Tableau

Power BI

## 🎯 Outcome

Job-ready Data Analytics foundation with a real-time project

1

**Excel Fundamentals**

2

**SQL for Data Analysis**

3

**Tableau Visualization**

4

**Power BI Analytics**

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**Real-Time Project**

# Deliverables for Students

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**Weekly assessments**



**One capstone project**



**Dashboards**

(Tableau & Power BI)



**SQL query practice**

# Month 1: Excel Fundamentals (Weeks 1–4)

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1

## Week 1

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- ✓ Intro to Data Analytics; Excel interface; data types
- ✓ Basic formulas: SUM, AVERAGE, COUNT, MIN, MAX
- ✓ Cell referencing (Relative, Absolute, Mixed)

2

## Week 2

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- ✓ Text functions (LEFT, RIGHT, MID, LEN, TRIM, CONCAT)
- ✓ Date & Time; Logic (IF, AND, OR)
- ✓ Sorting/Filtering; Data validation; Conditional formatting

3

## Week 3

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- ✓ Lookup (VLOOKUP, HLOOKUP, XLOOKUP, INDEX & MATCH)
- ✓ Pivot Tables/Charts; Removing duplicates
- ✓ Data cleaning; Basic dashboards

4

## Week 4

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- ✓ Advanced Pivot Tables; Slicers & Timelines
- ✓ What-if Analysis; Business datasets
- ✓ Mini case study using Excel

# Month 1: Weekly Assessments

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## TIMELINE

## ASSESSMENT FOCUS

### Week 1

📅 Friday Assessment

MCQs + Excel basics practical

### Week 2

📅 Friday Assessment

Excel functions practical test

### Week 3

📅 Friday Assessment

Pivot table & lookup-based task

### Week 4

📅 Friday Assessment

Excel case study

# Month 2: SQL for Data Analysis (Weeks 5–8)

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## 5 SQL Basics

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Databases & Tables concepts

SQL syntax & keywords

SELECT, DISTINCT

WHERE clause

ORDER BY

LIMIT

## 7 Joins & Subqueries

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INNER, LEFT, RIGHT, FULL JOIN

Joining multiple tables

Subqueries

Nested queries

## 6 Aggregations

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COUNT, SUM, AVG, MIN, MAX

GROUP BY

HAVING

Aliases

Filtering data

## 8 Advanced SQL

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CASE WHEN

Date functions

String functions

Window functions (ROW\_NUMBER, RANK)

SQL case study

# Month 2: Weekly Assessments

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## Week 5

📅 Friday Assessment

SQL basic queries



## Week 6

📅 Friday Assessment

Aggregation-based SQL test



## Week 7

📅 Friday Assessment

Join & subquery practical



## Week 8

📅 Friday Assessment

SQL case study





# Month 3: Tableau (Weeks 9–10)

## Week 9

- ✓ Tableau basics
- ✓ Connect data sources
- ✓ Dimensions & Measures
- ✓ Charts: Bar, Line, Pie, Scatter
- ✓ Filters & Sorting

## Week 10

- ✓ Calculated fields
- ✓ Parameters
- ✓ Table calculations
- ✓ Dashboards & Stories
- ✓ Tableau case study

### Friday Assessments

Weekly Evaluation

W9

Tableau charts assignment

W10

Tableau dashboard

# Month 3: Power BI (Weeks 11–12)



## Week 11

Basics & Visualization

- ✓ Power BI basics
- ✓ Power BI Desktop interface
- ✓ Data loading & transformation
- ✓ Power Query basics
- ✓ Basic visualizations



FRIDAY ASSESSMENT

Power BI visuals



## Week 12

Modeling & DAX

- ✓ Data modeling
- ✓ Relationships
- ✓ DAX basics: SUM, CALCULATE, FILTER
- ✓ Measures vs Columns
- ✓ Power BI dashboard creation



FRIDAY ASSESSMENT

Power BI dashboard

# Month 4: Real-Time Project & Interview Preparation (Weeks 13–16)

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# Assessments & Final Evaluation

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## Weekly Assessment

Every Friday



## Final Evaluation

Project + Viva